

Minsung Cho

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Software engineer working on compilers, database engines, and distributed systems.

Employment

Software Engineer, Compiler

May 2025 – Present

RelationalAI, Logic Engine team

Remote

- Design and implement the procedural extension to RelationalAI’s query language
 - Scale adoption from internal algorithms to several enterprise workloads (15k queries/day)
 - Cut day-long query runtimes to under an hour
- Lead migration of legacy customer code (>300k LoC) to our Python library, partnering with GTM
 - Build a compiler toolchain to incrementally transpile legacy code
 - Accelerate benchmark workloads up to 30% by eliminating costly legacy compiler passes
- Ship GQL-style graph pattern matching as sole compiler engineer with the Symbolic AI team
- Maintain the database engine’s Protocol Buffer-based entry point (100k+ queries/day)

Graduate Researcher

Sep 2022 – Apr 2025

Northeastern University, Programming Research Lab

Boston, USA

- Designed a language to express optimization problems under uncertainty [1]
 - Implemented a novel binary decision diagram algorithm to compute optimal values
 - Outperformed a state-of-the-art probabilistic programming language by up to 10x
- Co-led the first empirical study of online proof assistant communities [2]

R&D Intern

May 2024 – Dec 2024

RelationalAI, Logic Engine team

Remote

- Shipped a decompiler that lifts optimized Datalog back to the source query
- Integrated it into the production query profiler to automatically flag performance bottlenecks

Education

Master of Science, Computer Science

Sep 2022 – May 2025

Northeastern University

Boston, USA

Bachelor of Science, Mathematics and Philosophy

Aug 2018 – May 2022

Carnegie Mellon University

Pittsburgh, USA

Skills

Programming languages: Julia, Python, Datalog, OCaml, Rust, Go, C/C++, Haskell, Standard ML

Tools: Unix, Git, Docker, Datadog, Observe, Snowflake, LLVM/MLIR

Awards and Recognition

First Place, ACM Student Research Competition at PLDI (top PL conference) **2023**

Scholarship Recipient, Programming Languages Mentoring Workshop (PLMW) at PLDI **2023**

University and College Honors, Carnegie Mellon University **2022**

Publications

- [1] Minsung Cho, John Gouwar, and Steven Holtzen. “Scaling Optimization over Uncertainty via Compilation”. In: *Proc. ACM Program. Lang. (OOPSLA)* (2025). DOI: 10.1145/3720500.
- [2] Gwenth Lincroft*, Minsung Cho*, et al. “Thirty-Three Years of Mathematicians and Software Engineers: A Case Study of Domain Expertise and Participation in Proof Assistant Ecosystems”. In: *21st IEEE/ACM International Conference on Mining Software Repositories, MSR 2024*. ACM, 2024, pp. 1–13. DOI: 10.1145/3643991.3644908. *Joint first authors.